

Turkish News Broadcast Lip Reading Spotting Benchmark and Baseline Development — 12-Month Research Plan

Project Vision

This research aims to construct a two-tiered (word-level + sentence-level) lip reading dataset from Turkish news broadcast videos on YouTube using an automatic annotation pipeline, and to develop baseline models for **visual word spotting** on this dataset. The resulting benchmark will be the first publicly available spotting-focused lip reading resource for Turkish, a morphologically rich and under-resourced language in the visual speech recognition literature.

Background and Motivation

Visual speech recognition (lip reading) has seen remarkable progress in well-resourced languages, driven by large-scale benchmarks such as LRW [arxiv.org/abs/1611.01599], LRS2/LRS3 [arxiv.org/abs/1809.00496], and LRW-1000 [arxiv.org/abs/1810.06990]. However, Turkish remains severely under-resourced: existing Turkish lip reading datasets contain at most a few thousand clips with limited vocabulary [Berkol & Sivri]. Furthermore, no Turkish dataset has been designed with **sign or word spotting** as a primary task. News broadcasts are an ideal source as anchors speak clearly toward the camera in standard Turkish, following the well-established methodology of LRS2, LRS3, and BSL-1K [arxiv.org/abs/2007.12131].

Phase 1 — Data Collection and ASR Pipeline (Months 1–3)

Objectives

Collect raw video data and produce word- and sentence-level time-aligned transcripts using Whisper ASR.

Video Collection

- **Sources:** YouTube channels of major Turkish broadcasters — TRT Haber, CNN Türk, NTV, Habertürk
- **Tool:** `yt-dlp` for automated downloading
- **Inclusion criteria:** Single-anchor shots where the presenter faces the camera directly; minimum resolution 720p; no graphic overlays or split-screen segments
- **Target:** Minimum 25 hours per channel → ~100+ hours of raw video

ASR-Based Label Generation

- `whisper-large-v3` [github.com/openai/whisper] is run with `word_timestamps=True`, yielding per-word start/end times at ~100ms precision
- Whisper has demonstrated strong Turkish ASR performance (WER 4.3–14.2%), further reducible via LoRA fine-tuning
- Each word is stored as `{surface_form, lemma, start_sec, end_sec, confidence}`
- Sentence segments are stored as `{text, start_sec, end_sec}`
- Words with confidence < 0.7 are flagged for manual review

Output: Time-aligned transcript JSON files per video segment

Phase 2 — Visual Processing and Clip Generation (Months 2–5)

(Partially parallel with Phase 1)

Face Detection and Lip Extraction Pipeline

1. **Face detection:** MediaPipe Face Mesh [google.github.io/mediapipe] for per-frame bounding box and 468-point landmark extraction
2. **Face tracking:** Re-verification every 15 frames; segments with tracking failure are discarded
3. **Lip region extraction:** Cropped using the 20 landmarks surrounding the mouth region; normalized to **88×88 px grayscale** (following LRW convention [arxiv.org/abs/1611.01599])
4. **Audio-visual synchronization check:** SyncNet [arxiv.org/abs/1611.05557] score computed per clip; clips below threshold (score < 3) are discarded

Word-Level Clip Generation

- Each Whisper word timestamp is extended by ± 15 frames (~ 0.6 s at 25fps) to capture coarticulation context
- This yields clips of approximately 0.6–1.2 s — consistent with LRW's fixed 29-frame standard [arxiv.org/abs/1611.01599] while allowing natural duration variance as in LRW-1000 [arxiv.org/abs/1810.06990]

Sentence-Level Clip Generation

- Whisper sentence segments between 1–8 s are retained as full clips
- Longer sentences (> 8 s) are split at natural pause boundaries
- Each clip is labeled with its full transcript

Output: Two separate clip pools — `word_clips/` and `sentence_clips/`

Phase 3 — Dataset Compilation and Benchmark Design (Months 4–7)

Vocabulary Selection

- Frequency analysis across all transcripts
- Inclusion threshold: minimum **300 training examples per class**
- **Lemma-based grouping** is evaluated as an option: due to Turkish agglutinative morphology, surface forms such as *ekonomi*, *ekonomide*, *ekonominin* are grouped under the same root — this is an original contribution specific to Turkish
- **Target:** 500+ word classes, 150,000+ word-level clips total

Benchmark Splits

Split	Ratio	Constraint
Train	%80	—
Validation	%10	Speaker-independent
Test	%10	Speaker- and channel-independent

The spotting evaluation set consists **exclusively of sentence-level clips**; word-level clips are reserved for model training, following the methodology of BSL-1K [arxiv.org/abs/2007.12131].

Manual Verification

- 5% of all clips sampled randomly and verified via Label Studio [labelstud.io]
- Annotation task distributed among student groups
- Inter-annotator agreement reported as Cohen's κ

Spotting Ground Truth Annotation

- For each sentence clip, target word occurrences are annotated with precise temporal boundaries (Whisper output + manual correction)
- This constitutes the ground truth for spotting evaluation — analogous to the approach used in the ECCV 2022 Sign Spotting Challenge [arxiv.org/abs/2211.09448]

Phase 4 — Baseline Model Training (Months 6–10)

(Partially parallel with Phase 3)

Stage A: Isolated Word Classifier (Embedding Extractor for Spotting)

The following models are trained sequentially on the word-level clip pool and evaluated on Top-1/Top-5 word classification accuracy. Each model serves a distinct role in establishing the performance envelope of the benchmark.

Model	Type	Role	Reference
3D CNN + BiGRU	CNN + RNN	Lower-bound baseline; fast to train, well-understood behavior on LRW-1000	[arxiv.org/abs/1810.06990]
ResNet-18 + MS-TCN	CNN + Temporal Conv	Strong CNN baseline; multi-scale temporal modeling, competitive on LRW	[arxiv.org/abs/2108.02456]
VTP / SwinLip	Transformer Encoder	Transformer baseline; replaces CNN visual encoder with Swin Transformer for improved spatial modeling at lower compute cost	VTP: [arxiv.org/abs/2204.02774] — SwinLip: [arxiv.org/abs/2505.04394]
AVKT	Audio-Visual Keyword Transformer	Spotting-specific baseline; directly addresses unconstrained keyword spotting in sentence-level clips via cross-modal attention	[doi.org/10.1049/cit2.12212]
AV-HuBERT + LoRA	Large Pretrained Model	Upper-bound reference; frozen backbone fine-tuned with Low-Rank Adaptation to minimize GPU requirements	AV-HuBERT: [arxiv.org/abs/2208.02456] — LoRA: [arxiv.org/abs/2106.09685]

Reported metrics: **Top-1 / Top-5 accuracy** on the isolated word classification task.

Stage B: Visual Word Spotting Model

Each embedding extractor trained in Stage A is subsequently evaluated on the spotting task using a **sliding window** inference procedure over sentence-level clips:

1. A window of fixed temporal length (e.g., 29 frames) slides over each sentence clip with a stride of 4 frames
2. Per-window embeddings are computed using the Stage A model
3. Cosine similarity or softmax scores are computed against target word class embeddings
4. Windows exceeding a learned score threshold are marked as candidate spotting regions
5. **Non-Maximum Suppression (NMS)** merges overlapping candidates

This formulation directly mirrors the pipeline established in BSL-1K [arxiv.org/abs/2007.12131] for sign spotting, adapted here for visual word spotting in Turkish. The AVKT model [doi.org/10.1049/cit2.12212] additionally incorporates audio-visual cross-attention at the spotting stage itself, offering a direct end-to-end spotting baseline beyond the sliding window approach.

Reported metrics: **Average Precision (AP)**, **ROC-AUC**, Precision/Recall at IoU thresholds of 0.25, 0.50, and 0.75.

Stage C: Ablation Studies

- Effect of training set size (25%, 50%, 75%, 100% of word clips) on spotting AP
- Effect of temporal context window size (± 5 , ± 10 , ± 15 , ± 20 frames) on word classification and spotting
- Surface-form labeling vs. lemma-based labeling for morphologically rich Turkish vocabulary
- Impact of AV sync filtering threshold (SyncNet score) on downstream model quality
- Visual-only vs. audio-visual models: quantifying the contribution of the visual modality in a clean broadcast speech setting, following the methodology of [arxiv.org/abs/2412.17129]

Phase 5 — Analysis, Dissemination, and Documentation (Months 10–12)

Error Analysis

- Performance breakdown by speaker head pose (yaw/pitch angle) — identified as the most critical degradation factor across all lip reading models [arxiv.org/abs/1810.06990]
- Error distribution by word length and syllable count (Turkish words are longer on average)
- Most confused word pairs (visual homophones in Turkish)
- Performance variance across channels and speakers

Public Release

What will be released:

- Preprocessed **grayscale cropped lip region images** (88×88 px), normalized with global mean and variance — following the LRW-1000 release convention
- Per-clip metadata: {word/sentence label, lemma, start_sec, end_sec, speaker_id, channel, split}
- Speaker-independent train/val/test split files
- Word frequency statistics and vocabulary list (surface form + lemma mapping)
- Full pipeline code (video download, ASR, face detection, lip extraction, sync verification) on **GitHub** under MIT license

What will NOT be released:

- Raw broadcast video footage
- Full-face or upper-body video frames
- Audio recordings

Rationale:

Releasing only the mouth RoI crops — rather than raw video or full-face images — serves two purposes: (1) it eliminates copyright concerns arising from YouTube's Terms of Service and broadcaster intellectual property rights, and (2) it removes the preprocessing burden for downstream researchers, enabling direct model training without replication of the full pipeline. This approach is consistent with established practice in the field, as adopted by LRW-1000 [arxiv.org/abs/1810.06990].

Target platform: HuggingFace Datasets [huggingface.co/datasets]

Target Publication Venues

- LREC-COLING, ACM Multimedia, ECCV/ICCV Workshop on Sign, Gesture and Activity

Timeline Overview

text

Month 1		Video collection + ASR pipeline setup
Month 2		ASR outputs + face detection kickoff
Month 3		Clip generation (word + sentence level)
Month 4		Dataset compilation + frequency analysis
Month 5		Manual verification + benchmark design
Month 6		3D CNN baseline training begins
Month 7		Dataset finalized + spotting ground truth
Month 8		ResNet + MS-TCN training
Month 9		Sliding window spotting model
Month 10		AV-HuBERT fine-tuning + ablations
Month 11		Error analysis + paper writing
Month 12		Revision + public release + presentation

Student Task Allocation (Suggested)

Group	Size	Responsibilities
A	2	Video downloading, ASR pipeline, quality filtering
B	2	Face detection, lip extraction, sync verification
C	2	Manual annotation, dataset statistics, benchmark design
D	2–3	Model training, spotting evaluation, ablation studies

Risks and Mitigations

Risk	Mitigation
Whisper transcription errors in fast/ accented speech	Confidence score filtering + manual spot-check sampling

Turkish morphology inflating vocabulary	Lemma-based grouping evaluated as alternative labeling strategy
Imbalanced word frequency distribution	Oversample rare classes; report per-class AP in spotting evaluation
GPU resource constraints for AV-HuBERT	Use frozen backbone + lightweight adapter (LoRA) [arxiv.org/abs/2106.09685]